L15-16: Quantum Computing Primitives

Vishvendra S. Poonia
Department of Electronics and Communication Engineering
Indian Institute of Technology Roorkee
vishvendra@ece.iitr.ac.in
vishvendra.punia@gmail.com

This session (15-16)

Purity and Coherence — Summary

Overview of a Quantum Algorithm (Shor's Algorithm)

Entanglement

Quantum Gates

Qiskit

Purity and Coherence

Coherence means "having phase relations between basis states."

- Consider the state: $|\psi\rangle = |0\rangle$
- The density matrix is:

$$\rho = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

- Fix a reference orthonormal basis $\{|i\rangle\}$. A state is coherent in a given basis if its density matrix has non-zero off-diagonal elements in that basis.
- In the computational basis: $\rho_{01} = 0$. Therefore, no coherence in the computational basis.
- But, if we check the purity: $Tr(\rho^2) = 1$
- Therefore, the state is pure but not coherent in the computational basis.
- But, the same state in x-basis:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle_x + |-\rangle_x)$$

- The density matrix in x basis does have off-diagonal elements and hence coherent.

Purity is basis-independent. Coherence is basis-dependent.

Purity (Basis-Independent Property)

- Let ρ be a density operator on Hilbert space H.

ρ is pure $\Leftrightarrow \rho^2 = \rho$

- Equivalent condition: $Tr(\rho^2) = 1$
- If, $Tr(\rho^2) < 1 \Longrightarrow$ the state is mixed.
- Measure of purity: $\gamma(\rho) = Tr(\rho^2)$
- Purity is:
  - Basis independent
  - Invariant under unitary transformations
  - Global property of the state

Purity tells us how mixed the state is but not the origin of the mixedness.

Mixedness can come from: (i) Classical probability (ii) Entanglement with environment, (iii) Decoherence processes

It measures how much information about the system is inaccessible to us.

Entanglement

$$\rho = \rho_A \otimes \rho_B$$

Entanglement

- **Let's consider two spin-1/2 systems:** $H_{Total} = C^2 \otimes C^2$
- The dimension of a 2-qubit system is: 4.
- The separable state is: $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$
- E.g. $|0\rangle \otimes |1\rangle$
- **Density Matrix factorizes as:**
- **The Bell State**
- **A bipartite pure state is entangled iff it cannot be written as:** $|\psi\rangle = |\phi\rangle \otimes |\chi\rangle$

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |u_i\rangle \otimes |v_i\rangle$$

Schmidt Decomposition

Any bipartite pure state:

With:

$$\lambda_i > 0$$
$$\sum \lambda_i^2 = 1$$

- **If r=1 → separable**
- **If r>1 → entangled**
- **For the singlet state:**

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

- $\lambda_1 = \lambda_2 = 1/\sqrt{2}$
- **Entanglement $\Longrightarrow$ global purity with local mixedness.**
- Single-qubit coherence lives in off-diagonal terms
- **Entanglement lives in non-factorizability of the density matrix**

Qubit: Summary

Qubit: The Physical System

- A qubit is a two-level quantum system.
- **Examples:**
  - Spin-½ particle (electron, nucleus)
  - Polarization of a photon
  - Two-level atom
- **Mathematically, its state lives in a 2D complex Hilbert space:** $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

with normalization: $|\alpha|^2 + |\beta|^2 = 1$

Computational Basis:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

- These form an orthonormal basis.

Key Properties:

- Superposition
  - State exists in linear combination.
- Global Phase: Physically indistinguishable:

$$|\psi\rangle \sim e^{i\theta}|\psi\rangle$$

- **Bloch Sphere Representation**
- Multi-Qubit Systems: Tensor product structure: $|\psi\rangle \in C^2 \otimes C^2$
- E.g.: $$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
- This is entangled (cannot be factorized).

Qudits: (D-dimensional quantum systems)

Generalization of the Qubits to d-dimensional space

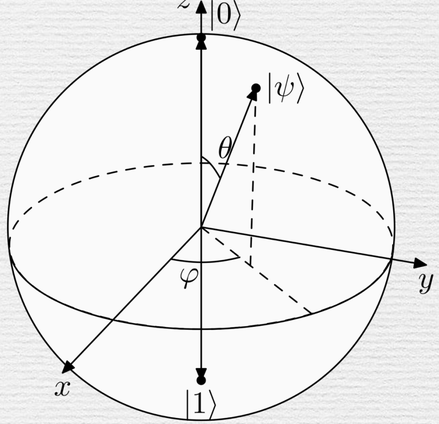
- with $$|\psi\rangle = \sum_{k=0}^{d-1} \alpha_k |k\rangle$$ $$\sum |\alpha_k|^2 = 1$$
- **Examples:**
  - Qutrit (d=3)
  - Photonic orbital angular momentum states
  - Higher spin systems

Why Qudits are important:

- Higher information density
- Native physical systems (ions, photons)
- Hilbert space dimension grows as:

$$(\mathbb{C}^d)^{\otimes n}$$

Bloch Sphere: Mixed States (Density Matrices)



We need density matrices when:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

- Entanglement (subsystems)
- Noise in the systems — Open Quantum Systems
- Statistical mixtures

Density Matrix: with properties:

- Hermitian
- Trace = 1
- Positive semi-definite

On Bloch Sphere: Every 2×2 density matrix can be written as: ; where and

The Bloch vector components are simply expectation values of Pauli operators:

$$\vec{r} = (r_x, r_y, r_z)$$

$$r_x = \langle\sigma_x\rangle, \quad r_y = \langle\sigma_y\rangle, \quad r_z = \langle\sigma_z\rangle$$

Important results:

- Pure states → surface (|r| = 1)
- Mixed states → inside sphere (|r| < 1)
- Maximally mixed state: → Center of sphere (r = 0)

$$\rho = \frac{I}{2}$$

Purity and Radius:

$$\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$$

- Purity: $\mathrm{Tr}(\rho^2) = \frac{1 + |\vec{r}|^2}{2}$
- So: |r| = 1 → purity = 1 (pure)
- |r| = 0 → purity = 1/2 (maximally mixed)

The length of the Bloch vector measures purity/coherence (to an extent).

Bloch Sphere: Key Takeaways

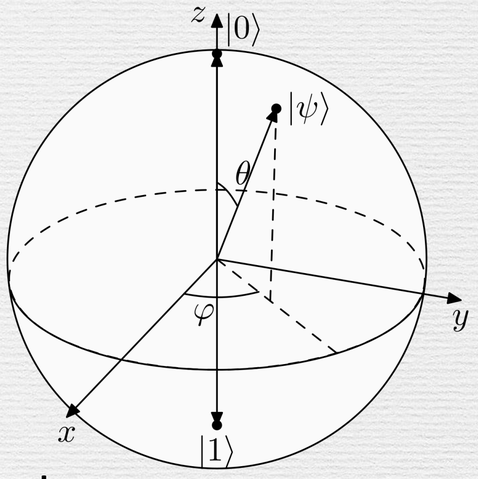
Unitary Evolution = Rotations

$$U = e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}}$$

- **Any single-qubit unitary: corresponds to:**
  - Rotation of Bloch vector around axis $\hat{n}$ by angle θ. $\hat{n} = (n_x, n_y, n_z)$ - Unit Vector. $\vec{\sigma} = (X, Y, Z)$
  - Also, $e^{i\theta(n\cdot\sigma)} = cos\theta.I + isin\theta.(n\cdot\sigma)$
- **Hence, Quantum gates = 3D rotations.**

Example:

- **Pauli X → 180° rotation about X-axis**
- **Pauli Y → 180° rotation about Y-axis**
- **Pauli Z → 180° rotation about Z-axis**
- Hadamard → 180° rotation about axis between X and Z



Decoherence

Shrinking of Bloch Vector

Noise channels:

- Depolarizing → shrinks uniformly
- Phase damping → shrinks X-Y components
- Amplitude damping → pulls toward $|0\rangle$

Geometrically: Noise = contraction + distortion of Bloch sphere.

Key Takeaways

- A qubit is not "0 or 1" — it's a direction in space.
- Global phase has no physical meaning.
- Pure states live on surface.
- Mixed states live inside.
- Unitary evolution = rotations. Noise = shrinking.

Quantum Algorithms

An Example: Shor's Algorithm

Shor's algorithm is a quantum algorithm developed by Peter Shor in 1994 that factors large integers exponentially faster than the best-known classical algorithms, specifically in polynomial time

Shor's Algorithm: Step-by-Step

To factor a large number , Shor's algorithm uses a "quantum heart" to find a period and a "classical shell" to turn that period into factors. Here is the step-by-step mathematical path from a random guess to the final prime factors:

- **Step 1: Select a Random Guess**
  - **Choose a random number a such that $1 < a < N$.**
  - **First, check if $gcd(a, N) > 1$ using the classical Euclidean algorithm.**
  - **If it is, we have accidentally found a factor! If not, proceed to the quantum step.**
- **Step 2: The Quantum Step [Find the Period (r)]**
  - **This is the only part that requires a QPU. We find the smallest positive integer r (the period) such that: $a^r \equiv 1(modN)$**
  - **This means that a^r is exactly one more than some multiple of N.**
- **Step 3: Verify the Period**
  - **For the math to work in the next step, the period must meet two conditions:**
    - **r must be even: If r is odd, the algorithm fails. Pick a new a and start over.**
    - **$a^{r/2} \not\equiv 1(modN)$: If this condition isn't met, the result will be trivial; pick a new a and start over.**
- **Step 4: Factor Using Difference of Squares**
  - **Since, r is even, we can rewrite the equation $a^r \equiv 1(modN)$ as: $a^r - 1 \equiv 0(modN)$**
  - **Using the algebraic identity for the difference of squares, we can split the term as: $(a^{r/2} - 1)(a^{r/2} + 1) = 0(modN)$**

Shor's Algorithm: Step-by-Step

- **Step 1: Select a Random Guess (a)**
- **Step 2: The Quantum Step [Find the Period (r)]**
- **Step 3: Verify the Period**
- **Step 4: Factor Using Difference of Squares**
  - **Since, r is even, we can rewrite the equation $a^r \equiv 1(modN)$ as: $a^r - 1 \equiv 0(modN)$**
  - **Using the algebraic identity for the difference of squares, we can split the term as: $(a^{r/2} - 1)(a^{r/2} + 1) = 0(modN)$**
  - **This tells us that the product $(a^{r/2} - 1)(a^{r/2} + 1)$ is a multiple of N. Therefore, the prime factors of must be distributed between these two terms.**
- **Step 5: Extract the factors**
  - **Use the classical Greatest Common Divisor (GCD) algorithm to find where the factors are hiding:**
  - **Factor 1: $gcd(a^{r/2} - 1, N)$**
  - **Factor 2: $gcd(a^{r/2} + 1, N)$**
- **Example: Factoring $N = 15$**
  - **Guess: Let $a = 7$.**
  - **Period: We find that $7^4 = 2401$. Since $2401 \equiv 1(mod15)$, the period is $r = 4$.**
  - **Split: r is even, so we calculate $7^{4/2} = 7^2 = 49$**
  - **Terms: The two terms are $49 - 1 = 48$ and $49 + 1 = 50$**
  - **Factors:**
    - $gcd(48,15) = 3$
    - $gcd(50,15) = 5$

By finding an even period r for the modular function, we can rewrite the congruence as a difference of squares, which allows us to extract the prime factors of using the Greatest Common Divisor of the resulting terms and N.

Shor’s Algorithm: Intuition

In Shor's algorithm, modular arithmetic is the mathematical backbone that allows us to find the period of a function, which is the key to factoring large numbers.

Here is why it is important:

- **Creates Periodicity: Shor’s algorithm relies on the function $f(x) = a^x (mod N)$. Because of the modulus N, the values eventually repeat in a cycle (the period r). Finding this period r allows us to calculate the factors of N .**

- **Restricts the Search Space: Without the modulus, a^x would simply grow toward infinity. Modular arithmetic keeps the values constrained within the range $[0, N-1]$, creating a finite structure that a Quantum Fourier Transform can analyze.**

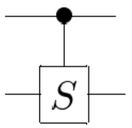
- **Enables Number Theory Tricks: The algorithm specifically uses Euler's Totient Theorem. If we find an even period r such that $a^r \equiv 1(mod N)$, we can rewrite this as $(a^{r/2} - 1)(a^{r/2} + 1) = kN$. This gives us the two numbers that likely share a greatest common divisor with N.**

Quantum Gates

Quantum gates = Unitary operators

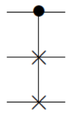
- **Linear**
- **Reversible**

controlled-phase

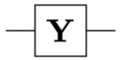
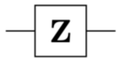
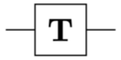
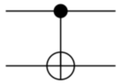
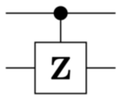
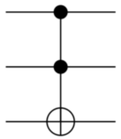


$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}$$

Fredkin (controlled-swap)



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Operator	Gate(s)		Matrix
Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)			$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)			$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)			$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)			$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Quantum Gates: Universal set of quantum gates

Quantum gates = Unitary operators

- **Linear**
- **Reversible**

- **Classical Universal: {AND, NOT} {NAND}, {NOR}**
- **A gate set G is universal if any unitary operator U can be approximated to arbitrary accuracy using a finite sequence of gates from G.**
- **A gate set becomes universal if it contains:**
  - **Arbitrary Single-Qubit Rotations** i.e. Any rotation on the Bloch sphere
  - **At Least One Entangling Two-Qubit Gate:** CNOT, CZ, iSWAP
- **Examples: {H,T,CNOT}, {Rx(θ),Rz(θ),CNOT}, {Toffoli,H}**
- **Geometrically, for single qubits, any point on Bloch sphere should be reachable from any other point.**
- **Similarly, two-qubit universality means that any entangled state can be produced**

Operator	Gate(s)		Matrix
Pauli-X (X)	X	$\oplus$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)	Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)	Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)	H		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)	S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)	T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)	$\bullet$ / $\oplus$		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)	$\bullet$ / Z	$\bullet$ / $\bullet$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP	$\times$	$\times$ / $\times$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)	$\bullet$ / $\bullet$ / $\oplus$		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Clifford gates: {H, S, CNOT}

Circuits with only Clifford gates (H, S,CNOT) can be efficiently simulated classically (Gottesman–Knill theorem) —> **In polynomial time**

To achieve universality we must add a non-Clifford gate such as T gate

Thank you